Unit 1 - Math Fundamentals
1.4 Series

Graham West

MSDS 520: Math and Stats Foundations for Data Science
Meharry Medical College

Outline

1. Why Series?
2. Sigma Notation
3. Finite Sums with Closed Forms
4. Infinite Series and Convergence
5. Back to the Dosing Problem
6. Review

Why Series?

A Problem to Motivate Us

A repeated dose

"A patient takes 100 mg of a drug every 6 hours. The drug's half-life is also 6 hours, so half of what is in the body is gone by the time the next dose arrives. Does the drug pile up forever?"

Track the amount in the body just after each dose:

$$\begin{aligned}
\text{after dose 1:} &\quad 100 \\
\text{after dose 2:} &\quad 100 + 100(0.5) \\
\text{after dose 3:} &\quad 100 + 100(0.5) + 100(0.5)^2 \\
\text{after dose } n: &\quad 100 + 100(0.5) + 100(0.5)^2 + \cdots + 100(0.5)^{n-1}
\end{aligned}$$

Every dose adds another term, so the sum has more in it every day, and yet the answer is **no**, the total does not grow forever. Understanding *why* is what this lecture is about.

Where This Shows Up in Data Science

You are already summing things constantly

- **Every statistic is a sum:** a mean, a variance, a total count.
- **Every loss function is a sum:** add up how wrong the model was on each of the n training rows.
- **Every dot product is a sum:** the core operation of Unit 2, and of every neural network.
- **Computers only add:** e^x, $\sin x$ and $\ln x$ are all evaluated as sums of infinitely many simple terms, cut off partway. (Unit 3 builds those sums.)
- **Training is a sequence of steps:** "does this algorithm settle down?" is exactly the question of whether a series converges.

The first three are the reason **sigma notation** is worth an hour: it is the language all of them are written in.

Unit 1-3 built **functions**. A series is what you get when you feed a function the numbers $1, 2, 3, \ldots$ and add up the results.

What We Will Build

The plan

1. **Sigma notation**: a compact way to write a long sum, and the rules for manipulating one. *This is the part you will use every week.*
2. **Closed forms**: finite sums that collapse to a single formula.
3. **Partial sums**: and what **convergence** means.
4. The **geometric series**, the one worth memorizing, and the answer to the dosing question.

Notebook for this lecture: `U1-4_Series-1_Convergence`.

Sigma Notation

Sequence First, Then Series

Sequence

A **sequence** is an ordered list of numbers, $a_1, a_2, a_3, \ldots$, one number for each index. It is a function whose domain is the counting numbers: feed it n, it returns a_n.

Series

A **series** is what you get when you **add up** a sequence:

$$S = a_1 + a_2 + a_3 + \cdots + a_n$$

The distinction is worth keeping straight: a sequence is the *list*, a series is the *total*. In a data table, a column of values is the sequence; the number at the bottom of the column is the series.

Writing a Long Sum Compactly

Writing out $a_1 + a_2 + \cdots + a_{100}$ is impractical, and continuing forever is impossible. **Sigma notation** says the same thing in one symbol:

$$\sum_{i=1}^{n} a_i \;=\; a_1 + a_2 + a_3 + \cdots + a_n$$

Reading the four parts

- $\sum$, the Greek capital sigma, meaning "add up what follows."
- i, the **index**: a counter that takes one value at a time.
- $i = 1$ and n, where the counter **starts** and **stops**, both included.
- a_i, the **general term**: the recipe for what is added, in terms of i.

Read it out loud as "the sum, as i runs from 1 to n, of a_i."

Shorthand. When the range is obvious or beside the point, the limits get left off: $\sum_i a_i$, or even $\sum a_i$. In statistics $\sum x_i$ means "over all n data points." Write the limits whenever there is doubt.

Sigma Is a For-Loop

The notation is exactly the loop you would write in code, start a counter, run to the limit, accumulate:

$$\sum_{i=1}^{4} i^2$$

$$= 1^2 + 2^2 + 3^2 + 4^2$$
$$= 1 + 4 + 9 + 16 = 30$$

```
total = 0
for i in range(1, 5):
    total += i**2
```

The upper limit is *inclusive* in mathematics and *exclusive* in Python's `range`, a mismatch that costs everyone a bug at least once.

The mechanical procedure

To expand a sigma: substitute the starting index into the general term, then the next value, and so on up to the upper limit, joining the results with plus signs.

The Index Is a Dummy Variable

The index has no meaning outside the sum, rename it and nothing changes:

$$\sum_{i=1}^{n} a_i \;=\; \sum_{j=1}^{n} a_j \;=\; \sum_{k=1}^{n} a_k$$

The starting value is a choice, not a law. These two write the same sum:

$$\underbrace{\sum_{i=1}^{n} 2^{i-1}}_{\text{start at } 1} \;=\; 1 + 2 + 4 + \cdots + 2^{n-1} \;=\; \underbrace{\sum_{i=0}^{n-1} 2^{i}}_{\text{start at } 0}$$

Shifting the start by one means shifting every appearance of i in the term to compensate. Starting at $i = 0$ is common for geometric series, because then the exponent *is* the index.

Rules for Manipulating Sums

Three rules do almost all the work. Each is just regrouping an ordinary addition:

Constants pull out

$$\sum_{i=1}^{n} c\,a_i \;=\; c\sum_{i=1}^{n} a_i \qquad \text{since} \qquad ca_1 + ca_2 + \cdots = c(a_1 + a_2 + \cdots)$$

Sums split

$$\sum_{i=1}^{n}(a_i + b_i) \;=\; \sum_{i=1}^{n} a_i \;+\; \sum_{i=1}^{n} b_i$$

A constant term counts itself

$$\sum_{i=1}^{n} c \;=\; \underbrace{c + c + \cdots + c}_{n \text{ times}} \;=\; nc$$

A Warning About What Does *Not* Split

The first two rules are called **linearity**, and linearity is about addition only. Products and powers do **not** pass through a sum:

$$\sum_{i=1}^{n} a_i b_i \;\neq\; \left(\sum_{i=1}^{n} a_i\right)\left(\sum_{i=1}^{n} b_i\right) \qquad \sum_{i=1}^{n} a_i^2 \;\neq\; \left(\sum_{i=1}^{n} a_i\right)^2$$

Check it with the smallest possible example

Take $n = 2$ with $a = (1, 1)$ and $b = (1, 1)$. Then $\sum a_i b_i = 1 + 1 = 2$, while $\left(\sum a_i\right)\left(\sum b_i\right) = 2 \cdot 2 = 4$.

That gap is not a nuisance, it is the entire content of the variance formula in Unit 4, where $\sum x_i^2$ and $\left(\sum x_i\right)^2$ appear side by side and their difference measures the spread of the data.

Exercise: Reading Sigma Notation

Try it

1. Write out every term of $\sum_{i=1}^{3} \frac{a_i}{b_i}$.
2. Write out every term of $\sum_{i=0}^{5} \frac{i^2}{i+1}$, and find the total.
3. Rewrite $3 + 6 + 9 + 12 + 15$ in sigma notation.
4. Simplify $\sum_{i=1}^{n}(4a_i + 7)$ using the rules on the previous slides.

Exercise: Solution

1 and 2

$\sum_{i=1}^{3} \frac{a_i}{b_i} = \frac{a_1}{b_1} + \frac{a_2}{b_2} + \frac{a_3}{b_3}$, the index rides along on *both* letters.

$\sum_{i=0}^{5} \frac{i^2}{i+1} = \frac{0}{1} + \frac{1}{2} + \frac{4}{3} + \frac{9}{4} + \frac{16}{5} + \frac{25}{6} = \frac{687}{60} = 11.45$, over a common denominator of 60. (Starting at $i = 0$ is legal, that term is simply zero.)

3 and 4

Each term is 3 times its position and there are five of them: $\sum_{i=1}^{5} 3i$. (Equally correct: $\sum_{i=0}^{4}(3i+3)$.)

Split, pull the constant out, count the sevens: $\sum_{i=1}^{n}(4a_i + 7) = 4\sum_{i=1}^{n} a_i + 7n$.

Sums You Already Use

Every one of these is sigma notation wearing a different name:

Four sums from the rest of this course

- **Mean** of n values: $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ (Unit 4)
- **Sum of squared errors:** $E = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ (Unit 3, optimization)
- **Dot product** of two lists: $\mathbf{v} \cdot \mathbf{w} = \sum_{i=1}^{p} v_i w_i$ (Unit 2, vectors)
- **Linear model** with p features: $\hat{y} = b + \sum_{j=1}^{p} w_j x_j$ (everywhere)

Once you can read a sigma you can read nearly every definition in machine learning, the notation is how the field writes its ideas down.

The Other Big Operator: $\prod$

The same idea with multiplication instead of addition uses a capital pi:

$$\prod_{i=1}^{n} a_i \;=\; a_1 \cdot a_2 \cdot a_3 \cdots a_n \qquad \text{for instance} \qquad \prod_{i=1}^{n} i \;=\; n!$$

Logarithms turn products into sums

$$\ln\left(\prod_{i=1}^{n} a_i\right) \;=\; \sum_{i=1}^{n} \ln a_i$$

This is not a curiosity. Multiplying thousands of probabilities together collapses to zero on a computer (Unit 1-5), so statistics works with the **log-likelihood**, the $\sum \ln$ on the right, instead of the product on the left. The logarithm rules from Unit 1-3 are what make the swap legal.

Finite Sums with Closed Forms

Gauss's Trick

The story

Set the busywork of adding 1 through 100, a young Gauss wrote the answer down immediately. He had noticed that the list can be folded in half.

Write the sum forward, then backward underneath, and add column by column:

$$\begin{aligned} S &= 1 \ + \ 2 \ + \ 3 \ + \cdots + \ n \\ S &= n + (n-1) + (n-2) + \cdots + \ 1 \end{aligned}$$

$$2S = \underbrace{(n+1) + (n+1) + \cdots + (n+1)}_{n \text{ columns, each equal to } n+1}$$

So $2S = n(n+1)$, giving the **closed form**

$$\sum_{i=1}^{n} i \ = \ \frac{n(n+1)}{2}$$

The Finite Geometric Sum

A **geometric** sequence multiplies by the same **ratio** r at every step: $a, ar, ar^2, ar^3, \ldots$ Its sum has a closed form too, found by a trick as slick as Gauss's.

Write the sum, multiply the whole thing by r, and subtract:

$$\begin{aligned} S_n &= a + ar + ar^2 + \cdots + ar^{n-1} \\ rS_n &= \quad ar + ar^2 + \cdots + ar^{n-1} + ar^n \end{aligned}$$

$$S_n - rS_n = a - ar^n$$

Everything in the middle cancels. Factor and divide:

$$S_n(1-r) = a(1-r^n) \qquad \Longrightarrow \qquad \boxed{\sum_{i=0}^{n-1} ar^i = a\,\frac{1-r^n}{1-r}} \quad (r \neq 1)$$

Infinite Series and Convergence

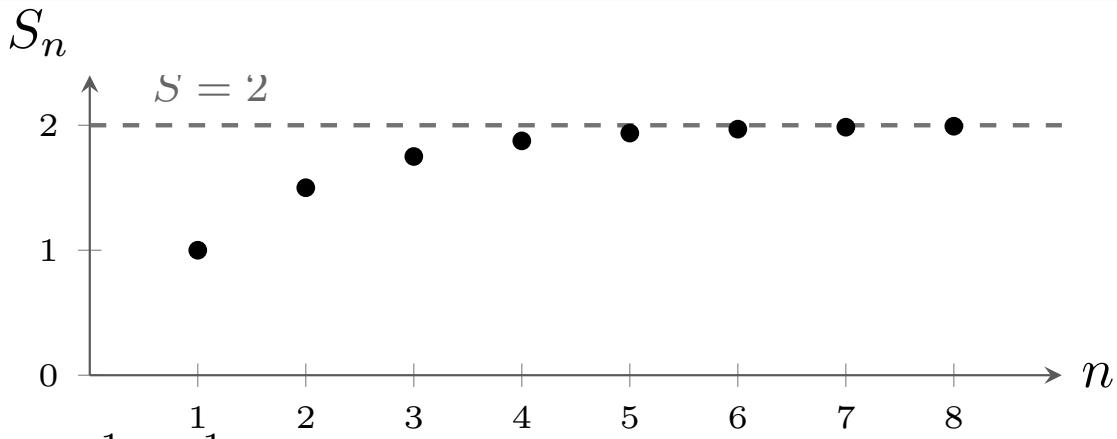
Partial Sums: How to Add Infinitely Many Things

We never actually add infinitely many numbers. We add the first n and watch what happens as n grows.

Partial sums and convergence

The nth **partial sum** is the total of the first n terms, $S_n = \sum_{i=1}^{n} a_i$, and these form a sequence of their own: $S_1, S_2, S_3, \ldots$

The series $\sum_{i=1}^{\infty} a_i$ **converges** to S if those partial sums settle on S, that is $S_n \to S$ as $n \to \infty$. If they settle on no number, it **diverges**.



Partial sums of $1 + \frac{1}{2} + \frac{1}{4} + \cdots$. Each term closes half the remaining gap to 2, so the dots approach the dashed line without ever reaching it. "The sum is 2" means exactly that: **an infinite sum is a limit of ordinary finite additions**, not an addition.

The Infinite Geometric Series

Start from the finite formula and let n grow. Everything hinges on the single term r^n: when $|r| < 1$, repeated multiplication by r drives $r^n \to 0$ and the formula collapses.

$$S_n = a\,\frac{1 - r^n}{1 - r} \qquad \Longrightarrow \qquad \boxed{\sum_{i=0}^{\infty} ar^i \;=\; \frac{a}{1 - r}} \quad \text{provided } |r| < 1$$

Ratio	Terms do what?	The series
$\lvert r\rvert < 1$	shrink toward 0	**converges** to $\dfrac{a}{1-r}$
$r = \pm 1$	stay the same size	diverges
$\lvert r\rvert > 1$	grow without bound	diverges

Worth memorizing: first term divided by one minus the ratio. **Always confirm** $|r| < 1$ **first**, since plugging in $r = 2$ returns $\frac{a}{-1} = -a$, a negative total for positive terms, which is the formula's way of saying it was never valid there.

One caution. Shrinking terms are *necessary* but not *sufficient*: the terms of $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$ shrink to zero, yet that sum grows without bound. Geometric shrinking is fast enough; merely shrinking is not.

Exercise: Geometric Series

Try it

Decide whether each series converges, and if it does, find its exact sum.

1. $\displaystyle\sum_{i=0}^{\infty} \frac{1}{2^i}$
2. $\displaystyle\sum_{i=1}^{\infty} \frac{1}{3^i}$
3. $\displaystyle\sum_{i=0}^{\infty} \left(\frac{3}{2}\right)^i$
4. $\displaystyle\sum_{i=0}^{\infty} 5\left(-\frac{1}{4}\right)^i$

For each one, identify a and r *first*, most mistakes here are bookkeeping mistakes about which term is the first one.

Exercise: Solution

1 and 2

(1) $a = 1$ (the $i = 0$ term) and $r = \frac{1}{2}$, so it converges to $\dfrac{1}{1 - 1/2} = 2$, the plot from a few slides back.

(2) This sum starts at $i = 1$, so the first term is $a = \frac{1}{3}$, not 1, with $r = \frac{1}{3}$: $\dfrac{1/3}{1 - 1/3} = \dfrac{1/3}{2/3} = \dfrac{1}{2}$.

3 and 4

(3) Here $r = \frac{3}{2} > 1$, so the terms grow: it **diverges**, and no sum exists. (The formula would have returned -2, nonsense, as promised.)

(4) $a = 5$ and $r = -\frac{1}{4}$, and $|r| = \frac{1}{4} < 1$, so it converges even though the terms alternate in sign: $\dfrac{5}{1 - (-1/4)} = \dfrac{5}{5/4} = 4$.

Back to the Dosing Problem

Answering the Opening Question

The setup, in this lecture's vocabulary

A dose of $D = 100$ mg every 6 hours, with a fraction $r = 0.5$ surviving until the next dose. Just after the nth dose the body holds

$$A_n \;=\; D + Dr + Dr^2 + \cdots + Dr^{\,n-1} \;=\; D\,\frac{1 - r^n}{1 - r}$$

, a **finite geometric sum**, exactly the one derived earlier.

Since $|r| < 1$, the term r^n dies away and the amount levels off at a **steady state**:

$$A_\infty \;=\; \frac{D}{1 - r} \;=\; \frac{100}{1 - 0.5} \;=\; 200 \text{ mg}$$

The drug does accumulate, but only to **twice** a single dose, never without bound. Each new dose is eventually offset by what the body clears, and the sum of infinitely many doses is finite.

Review

Review

What we covered

1. A **sequence** is a list; a **series** is its total. **Sigma notation** $\sum_{i=1}^{n} a_i$ is a for-loop written compactly: start, stop, and a rule for the term.
2. The index is a **dummy variable**, so its name never matters and never escapes the sum.
3. Sums are **linear**, constants pull out and sums split, but products and squares do *not* pass through: $\sum a_i b_i \neq (\sum a_i)(\sum b_i)$.
4. **Closed forms** replace a loop with a formula: $\sum i = \frac{n(n+1)}{2}$, and the geometric sum $a\frac{1-r^n}{1-r}$.
5. An infinite series is a **limit of partial sums**: it converges if S_n settles on a number, and diverges otherwise.
6. **Geometric series:** $\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r}$ when $|r| < 1$, the one to memorize, and the reason a repeated dose levels off instead of piling up forever.

Notebook: `U1-4_Series-1_Convergence`. **Previous:** Unit 1-3, Functions. **Next:** Unit 1-5, **Floating-Point Arithmetic**.